Group Project

SECP 3223 Data Analytics Programming

20242025 – Sem 2

PHASE I: Data Preparation and Cleaning Due Date: 12th June 2025

PHASE II: Data Analytics, Visualisation and Video Recording: 30th June 2025

DATA THEME: CO2 EMISSIONS FROM CARS

The rapid increase in vehicle usage has significantly contributed to global carbon dioxide (CO2) emissions, leading to severe environmental and health concerns. As the transportation sector remains one of the largest sources of greenhouse gases, understanding the impact of CO2 emissions from cars is critical for developing sustainable solutions. This data analytics project aims to analyze real-world vehicle emission data to **uncover patterns, identify high-emission vehicle categories, and propose strategies for reducing carbon footprints**. We will use Python-based analytics tools to explore fuel consumption trends and emission factors to gain insights into the relationship between vehicle characteristics and environmental impact.

You will be working on a data analytics project using datasets that can be downloaded from the e-Learning:

i) `FuelConsumption.csv`
   - This dataset consists of various car models, fuel consumption and their carbon emissions recorded in 2014

ii) `Cars2025.csv`
   - This dataset consists of car registrations in Malaysia for the year 2025

Instructions:

1. This group project has a maximum of 5 students per group.
2. This project is to be conducted in **TWO PHASES**:
   a. **Phase I: DATA PREPARATION AND CLEANING**
   b. **Phase II: DATA ANALYTICS, VISUALIZATION AND VIDEO RECORDING**

2. This project aims to expose students to the data analytics process starting from data preparation, data wrangling, visualization and machine learning application.
3. This project is divided into several tasks:

Phase I: DATA PREPARATION AND CLEANING

i. Prepare both datasets and match the car registration from **`Cars2025.csv`** with the information on the carbon emissions from **`FuelConsumption.csv`**
ii. Do appropriate data cleaning, preparation and wrangling.
iii. Do appropriate data aggregation and group operations.

Phase II: DATA ANALYTICS, VISUALIZATION AND VIDEO RECORDING

i. Perform summary analytics to highlight important information from the prepared dataset.
ii. Perform appropriate machine learning method to classify / cluster the car maker / car model according to their carbon emissions using machine learning.
iii. Visualize your analysis using appropriate visualization.

4. The submission of the project will be via e-learning. Documents to submit:
   a. Python codes and report in Jupyter Notebook (ipynb format) which explain in detail all the tasks in step 3.
   b. Initial dataset (before performing any operations)
   c. Video presentation (any software of your choice) – the duration is **max 10 minutes**.

- Please zip items a – b and upload in e-learning
- Item c to be uploaded in your Google Drive and the link will be shared in your Jupyter Notebook (item a). Please make sure that the video can be viewed by the lecturer.

5. Please refer to the rubric given to see the assessment method of this project.

Please make sure that all group members' details (name and matric number) are written in all submitted documents (Jupyter Notebook file and video presentation).

MARK SHEET FOR GROUP PROJECT

ITEMS	MARKS		
	1 - 3	**4 - 7**	**8 - 10**
PART A – REPORT ***Section i***			
Dataset			
Research Question			
Section Total	(/ 20 × 5) =		
Section ii			
Data Cleaning and Preparation			
Data Aggregation and Group Operations			
Analysis and visualization			
Machine Learning			
Section Total	(/ 40 × 25) =		
Section iii			
Conclusion			
Overall Writing			
Section Total	(/ 20 × 5) =		
TOTAL (PART A)			
PART B – ipynb PYTHON CODE			
Code Review			
TOTAL (PART B)	(/ 10 × 5) =		
PART C – VIDEO PRESENTATION			
Verbal Presentation			
Visual Presentation			
Overall Presentation			
TOTAL (PART C)	(/ 30 × 10) =		
OVERALL TOTAL	**(/ 50 × 25) =**		

RUBRIC FOR GROUP PROJECT

ITEMS	MARKS		
	1 - 3	**4 - 7**	**8 - 10**
PART A - REPORT			
Dataset	Too simple with not enough data to perform data preparation and analysis	Appropriate with enough data to perform data preparation and analysis	Good choice of dataset with large amount of data which can be used to perform complete data preparation and analysis
Research Question	Questions overly simplistic, unrelated, or unmotivated	Questions appropriate, coherent, and motivated	Questions well motivated, interesting, insightful, and novel
Data Cleaning and Preparation	Overly simplistic or incomplete	Appropriate and complete	Appropriate, complete and advanced
Data Aggregation and Group Operations	Overly simplistic or incomplete	Appropriate and complete	Appropriate, complete and advanced
Analysis and visualization	Choice of analysis is overly simplistic or incomplete. Inappropriate choice of plots; poorly labelled plots; plots missing	Analysis appropriate. Plots convey information but lack context for interpretation	Analysis appropriate, complete, advanced, and informative. Plots convey information correctly with adequate and appropriate reference information
Machine Learning	Overly simplistic or incomplete	Appropriate and complete	Appropriate, complete and advanced
Conclusion	Conclusions are missing, incorrect, or not based on analysis	Conclusions relevant, but partially correct or partially complete	Relevant conclusions explicitly tied to analysis and to context
Overall Writing	Explanation is illogical, incorrect, or incoherent	Explanation is correct, complete, and convincing	Explanation is correct, complete, convincing, and elegant
PART B – ipynb PYTHON CODE			
Code Review	Code is messy and poorly organized; unused or irrelevant code distracts when reading code. Variables and functions names do not helpful to understand code.	Code is reasonably well organized. There is little unused or irrelevant code, or this code has been moved out of the main project files. Variable and function names	Code very well organized. No irrelevant or distracting code. Variable and function names have clear relationship to their purpose in the

		generally meaningful and helpful for understanding.	code. Code is easy to read and understand.
PART C – VIDEO PRESENTATION			
Verbal Presentation	Illogical, incorrect, or incoherent.	Partially correct but incomplete or unconvincing	Correct, complete, and convincing
Visual Presentation	Cluttered, disjoint, or illegible	Readable and clear	Appealing, informative, and crisp
Overall Presentation	Verbal and visual presentation unrelated	Verbal and visual presentation related	Verbal and visual presentation clearly related